Sanskar Muneshwar Velocity Bikes Analysis

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-- Velocity Bikes Ltd. - SQL Capstone Project (Day 3 Deliverable)
-- Analyst: Sanskar Muneshwar
-- Database: bike_store_db
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USE bike_store_db;

-- ---------------------------------------------------------------------
-- Question 1: Why are some products selling better than others?
-- Analyzing product price tiers against sales volume and total revenue.
-- ---------------------------------------------------------------------
SELECT
    CASE
        WHEN p.price < 500 THEN 'Budget (< $500)'
        WHEN p.price BETWEEN 500 AND 1500 THEN 'Mid-Range ($500 - $1,500)'
        WHEN p.price BETWEEN 1501 AND 3000 THEN 'High-End ($1,501 - $3,000)'
        ELSE 'Premium (> $3,000)'
    END AS price_segment,
    COUNT(DISTINCT p.product_id) AS product_count,
    SUM(o.quantity) AS total_units_sold,
    ROUND(SUM(o.quantity * o.price), 2) AS total_revenue
FROM products p
LEFT JOIN orders o ON p.product_id = o.product_id
GROUP BY price_segment
ORDER BY total_revenue DESC;


-- ---------------------------------------------------------------------
-- Question 2: Which customers could be considered for a loyalty programme?
-- Finding top spenders with frequent repeat purchases.
-- ---------------------------------------------------------------------
SELECT
    c.customer_id,
    c.first_name,
    c.last_name,
    c.email,
    c.state,
    COUNT(DISTINCT o.order_id) AS total_orders,
    SUM(o.quantity) AS total_items_bought,
    ROUND(SUM(o.quantity * o.price), 2) AS total_spend
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.first_name, c.last_name, c.email, c.state
HAVING total_spend > 25000 AND total_orders >= 2
ORDER BY total_spend DESC
LIMIT 10;


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-- Question 3: Are there states with many customers but low revenue?
-- Analyzing customer concentration vs. revenue generated per customer.
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SELECT
    c.state,
    COUNT(DISTINCT c.customer_id) AS total_customers,
    COUNT(DISTINCT o.order_id) AS total_orders,
    ROUND(SUM(o.quantity * o.price), 2) AS total_revenue,
    ROUND(SUM(o.quantity * o.price) / COUNT(DISTINCT c.customer_id), 2) AS
revenue_per_customer
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.state
ORDER BY total_customers DESC;


-- -----------------------------------------------------------------------
-- Question 4 & 5: Low-performing products — Discount or discontinue?
-- Identifying unsold or dead inventory products.
-- -----------------------------------------------------------------------
SELECT
    p.product_id,
    p.product_name,
    p.model_year,
    p.price,
    COALESCE(SUM(o.quantity), 0) AS units_sold,
    COALESCE(ROUND(SUM(o.quantity * o.price), 2), 0.00) AS total_revenue
FROM products p
LEFT JOIN orders o ON p.product_id = o.product_id
GROUP BY p.product_id, p.product_name, p.model_year, p.price
HAVING units_sold = 0
ORDER BY p.model_year ASC, p.price DESC;


-- -----------------------------------------------------------------------
-- Question 6: Are newer bicycle models performing better than older ones?
-- Comparing performance metrics across model release years.
-- -----------------------------------------------------------------------
SELECT
    p.model_year,
    COUNT(DISTINCT p.product_id) AS total_models,
    COALESCE(SUM(o.quantity), 0) AS units_sold,
    ROUND(COALESCE(SUM(o.quantity * o.price), 0), 2) AS total_revenue,
    ROUND(COALESCE(SUM(o.quantity * o.price), 0) / COUNT(DISTINCT p.product_id), 2) AS
avg_revenue_per_model
FROM products p
LEFT JOIN orders o ON p.product_id = o.product_id
GROUP BY p.model_year
ORDER BY p.model_year ASC;


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-- Question 7: Which regions could be good opportunities for expansion?
-- City-level sales volume within underserved / growing states.
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SELECT
    c.state,
    c.city,
    COUNT(DISTINCT c.customer_id) AS customer_count,
    ROUND(SUM(o.quantity * o.price), 2) AS city_revenue
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.state, c.city
ORDER BY city_revenue DESC
LIMIT 10;
```

Part 2: Business Insights Report Blueprint

1. Executive Summary

- **Velocity Bikes Ltd.** generated a total lifetime revenue of **$8,578,988.88** across **1,615 orders** and **7,078 units sold**.
- While budget bikes drive high foot traffic and volume (2,915 units), premium bikes ($3,000+) account for the largest share of revenue ($3.19M, or 37.2% of total revenue).
- Geographic performance is heavily anchored in New York (67.9% of total revenue), but Texas exhibits the highest spend per customer ($6,778.88), making it a prime candidate for retail expansion.

2. Key Findings

- **Price Elasticity & Volume:**
  - Products priced under $500 drove **41.2% of sales volume** (2,915 units) but only 12.1% of revenue ($1.04M).
  - In contrast, high-end bikes ($1,500+) drive **66.3% of revenue** despite representing only 25.8% of volume.
- **Model Year Performance:**
  - **2016 models** generated $3.55M across just 25 products ($142,203 average per product).
  - **2017 models** generated $3.56M across 85 products ($41,941 average per product).
  - **2018 models** generated $1.46M across 197 products ($7,405 average per product).
  - *Insight:* Newer catalog additions suffered from catalog bloat and reduced average sales velocity per model.
- **Dead Stock Analysis:**
  - **14 products have zero recorded sales** (including older 2016 models like *Trek 820 - 2016* and various 2018 kids/cruiser bikes).
- **Loyalty Candidates:**
  - Top spenders like **Pamelia Newman** ($37,801.84), **Abby Gamble** ($37,500.89), and **Sharyn Hopkins** ($37,138.86) have placed multiple high-ticket orders and represent prime targets for high-tier perks.

3. SQL Outputs

- In your Word document, paste the SQL query code for each question followed by a screenshot of the output grid from MySQL Workbench.

4. Business Recommendations

- **Catalog Optimization:** Discontinue unsold 2016 and 2017 inventory. For unsold 2018 models (especially niche kids' bikes), offer clearance discounts of 15–20% to liquidate warehouse space.

- **Launch a Tiered Loyalty Program:** Offer VIP servicing, early access to new frame drops, and complimentary tune-ups to customers spending over $25,000.
- **Regional Expansion in Texas:** Texas customers have the highest revenue per customer ($6,778.88 vs. $5,717.61 in NY). Prioritize local marketing and service centers in high-revenue Texas cities.
- **Refocus Catalog Strategy on High-Margin Categories:** Rather than adding dozens of minor variations, prioritize high-performance road and trail bikes where margins are highest.